

INFT-1206-04

Assignment#4

Written Reports

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*No funding was received. Correspondence concerning this assignment should
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Project Overview

This report documents the design, development, and evaluation of two web-based projects completed for Assignment 4 of INFT1206: Web Development. Both projects were built using HTML5, CSS3, and Vanilla JavaScript with no external libraries or frameworks, and require no web server to run - each file opens directly in any modern browser.

The two deliverables are:

- **Part 1 - resume.html:** An interactive, animated resume page presenting professional experience, education, skills, certifications, projects, and contact information in a dark executive aesthetic.
- **Part 2 - dashboard.html:** An interactive skills dashboard that visually displays nine technology skills with animated progress indicators, card-flip detail views, category filtering, and animated statistics.

Both pages share the same visual design language: a near-black background, electric blue accent colour, monospace typography for labels, and serif typography for headings. This creates a cohesive, professional portfolio identity consistent across both deliverables.

Part 1: Interactive Resume

1.1 Description of the Resume Page

The page is structured as a continuous scrollable document divided into clearly labelled sections: Professional Profile, Experience, Education, Technical Skills, Credentials, Projects, Tools & Technologies, and Contact.

The page opens with a boot sequence loading animation that simulates a system initialisation before revealing the main content. A sticky navigation bar at the top allows the user to jump directly to any section. The navigation link for the currently visible section is highlighted automatically as the user scrolls. All content is hardcoded directly in the HTML file, meaning no web server or file fetching is required - the file works by simply double-clicking it.

The personal identity mark - the typographic logo RAYYAN UMAIR rendered in large serif type with the surname in the accent colour - serves as the main visual identity element for the page, fulfilling the assignment requirement for a main image or personal logo.

1.2 Colour Scheme

Background: #08080f - near-black with a subtle blue undertone

Card background: #0e0e1a - slightly lighter than the base

Primary accent: #4f7cff - electric blue used for labels, highlights, and borders

Primary text: #e8e8f5 - warm off-white for high readability on dark background

Secondary text: #a0a0c0 - muted slate for descriptions and body copy

Muted text: #50506a - used for metadata, dates, and labels

Green: #00e676 - used for active/complete status indicators

Amber: #f5a623 - used for Support & Operations skill category

Cyan: #00d4ff - used for Security skill category

Purple: #b388ff - used for Tools & Scripting skill category

The colour palette was chosen to convey seriousness, technical authority, and premium quality. The near-black background reduces eye strain and creates strong contrast with the light text. The single electric-blue accent colour is used sparingly to draw attention without overwhelming the page. Category colours for skills provide visual differentiation without introducing visual noise.

1.3 Layout

The layout uses CSS Grid and Flexbox throughout, with no table-based layout anywhere in the file. The overall page structure is a centered single-column container (max-width: 1200px) with generous horizontal padding.

Specific layout choices per section:

- Hero header: CSS Grid (2 columns) - name/thesis block on the left, availability status on the right.
- Education: CSS Grid (2 equal columns) - one card per degree.
- Technical Skills: CSS Grid (2 columns) - one block per skill category.
- Certifications: CSS Grid (3 columns) - one card per certification.

- Projects: CSS Grid (2 columns) - one card per project.
- Contact: CSS Grid (4 columns) - one clickable card per contact link.
- Navigation: Flexbox row - links evenly spaced in a sticky bar.

All grids collapse to a single column on screens narrower than 768px, and the 2-column grids for certifications and contact collapse at 480px, ensuring the page is fully responsive across devices.

1.4 Interactive Features

The resume includes eight distinct interactive and animated elements, exceeding the assignment minimum of five:

1. Custom Cursor

The default browser cursor is hidden and replaced with a two-part custom cursor: a solid blue dot that snaps to the mouse position instantly, and a slightly larger ring that smoothly lags behind using linear interpolation (lerp) driven by `requestAnimationFrame`. Both elements scale up when hovering over interactive elements such as cards, navigation links, and contact cards.

2. Loading Animation (Boot Sequence)

When the page first loads, a full-screen overlay displays the name RAYYAN UMAIR in large serif type alongside an animated progress bar and a sequence of status messages: INITIALISING PROFILE, LOADING CREDENTIALS, MAPPING EXPERIENCE, COMPILING PROJECTS, and ACCESS GRANTED. Each step advances on a 380ms timer. When the final step completes, the loader fades out and the main page content fades in and slides upward into view.

3. Accordion Experience Cards

Each work experience card has a clickable header row. Clicking it expands the card to reveal a bulleted list of job responsibilities and a row of tool tags. Only one card can be open at a time - opening a new card automatically closes the previously open one. The expand/collapse animation is driven by CSS max-height transition. Keyboard users can also activate cards using Enter or Space, with full aria-expanded attribute management for screen reader accessibility.

4. Hover Lift on Cards

Education cards, certification cards, and project cards all lift 4px upward on hover with an accompanying blue glow box-shadow. This is achieved entirely through CSS transitions on transform: translateY and box-shadow, with no JavaScript required.

5. Scroll-Triggered Skill Bar Animation

Skill progress bars start at width 0%. An IntersectionObserver watches the skills section. When the section enters the viewport for the first time, each bar animates to its target width using a 1-second cubic-bezier ease transition. The observer disconnects after the first trigger so the animation only fires once per page load.

6. Scroll Reveal

Every major card and block on the page starts at opacity 0 and translateY(20px). As each element scrolls into view, an IntersectionObserver adds the .visible class, triggering a CSS fade-in and slide-up transition.

7. Active Navigation Link

An IntersectionObserver watches each section element. As sections enter the viewport during scrolling, the corresponding navigation link in the sticky bar has the .active class applied, highlighting it in the accent colour with an underline.

8. Animated Background Canvas

A fixed HTML5 Canvas element behind all page content displays a continuously animating network of floating nodes and fading connection lines. Nodes drift slowly and bounce off viewport edges. Lines between nearby nodes fade out with distance. The animation runs in a requestAnimationFrame loop and rebuilds on window resize.

1.5 Animation Choices

Animation decisions were made to serve the professional identity of the page rather than to entertain. The boot sequence establishes a technical, system-like tone from the first interaction. The scroll-triggered animations reward attentive reading by revealing content as it becomes relevant. The background canvas adds depth and field-connection without ever competing with the readable content due to its low opacity setting of 22%.

The card flip and hover lift effects were chosen specifically because they demonstrate interactivity without requiring navigation away from the page, keeping the user engaged with a single continuous document.

1.6 Progress Indicator Style

Skills are represented as thin horizontal progress bars (2px height) positioned below each skill name and context label within a labelled category block. Each bar starts at 0% width and animates to its target percentage when scrolled into view. Bars are colour-coded by category: amber for Support, cyan for Security, green for Systems, and purple for Tools & Scripting. The percentage value is also displayed as text to the right of the skill name, providing an exact figure alongside the visual indicator.


```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="utf-8">
5    <title>Rayyan Umair | Resume</title>
6    <link rel="stylesheet" href="css/resume.css">
7  </head>
8  <body>
9    <!-- loader placeholder -->
10   <div id="loader">
11     <p class="loader-name">RAYYAN</p>
12   </div>
13   <div id="page-wrap">
14     <!-- sections will go here -->
15   </div>
16   <script src="js/resume.js"></script>
17 </body>
18 </html>

```

Early Build – Basic HTML structure before styling or JS

RAYYAN

UMAIR in progress...

INITIALISING...

<!-- page structure not yet complete -->

Nav, sections, and JS not yet wired up.

Browser preview – Early stage: loader visible, layout not yet built

PROFILE EXPERIENCE EDUCATION SKILLS CREDENTIALS PROJECTS CONTACT

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RAYYAN UMAIR

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Oshawa, ON, Canada

Part 2 : Interactive Skills Dashboard

2.1 Description of the Dashboard

dashboard.html is a standalone interactive skills dashboard that presents nine technology skills in a three-column card grid. Each card displays the skill name, a short description, a category label, a percentage proficiency value, and an animated progress bar. The dashboard includes a filter bar at the top that lets users show only cards from a selected category. Clicking any card flips it over to reveal a detailed back face with a proficiency tier badge, context about where the skill was applied, related tools, and a next-steps goal.

Below the skill grid, a summary statistics bar displays aggregate data - total skills tracked, average proficiency, number of certifications, and years of experience - with each number counting up from zero when the bar enters the viewport. A colour legend below the stats bar identifies which colour corresponds to which skill category.

Like the resume page, the dashboard is fully self-contained and requires no server. It shares the same visual identity, cursor, and canvas background as resume.html.

2.2 Chosen Skills

Nine skills were selected to represent the full breadth of the professional skill set documented in the resume. They were chosen to cover all four major skill domains - Security, Networking, Systems, Scripting, and Support - ensuring a balanced and honest representation:

- CompTIA Security+ (Security, 82%) - The core security certification, applied in SIEM monitoring and vulnerability assessment work.
- SIEM Monitoring (Security, 75%) - Hands-on log review and alert triage experience at SecureLogix using Splunk.
- Penetration Testing (Security, 70%) - Supervised web and network assessments completed at Tool-Go Inc.
- Active Directory (Systems, 88%) - Daily use at SecureLogix for user account and access management.
- Linux CLI & Admin (Systems, 72%) - Used extensively in security labs and pen testing tools.
- Network Analysis (Networking, 76%) - Wireshark-based packet capture and protocol dissection in labs and internship.
- Python Scripting (Scripting, 68%) - Automation scripts for IT support tasks and security tooling.
- IT Support Tier 1 & 2 (Support, 92%) - Most developed skill across three professional roles since 2023.
- Vulnerability Scanning (Tools, 78%) - Risk assessments and remediation reporting using Nessus and Nmap.

Proficiency percentages are self-assessed based on depth of real-world application, not exam scores. They are honest representations rather than inflated figures.

2.3 Colour Scheme

The dashboard inherits the same base colour palette as the resume page. Category accent colours are applied consistently to both the card category label and the progress bar fill:

Security cards: #00d4ff (cyan)

Networking cards: #4f7cff (electric blue / accent)

Systems cards: #00e676 (green)

Scripting cards: #b388ff (purple)

Support cards: #f5a623 (amber)

Tools cards: #ff5370 (red)

The card back face uses a slightly lighter card background (#131325) to signal the state change from front to back, creating a clear visual distinction without introducing a jarring colour shift.

2.4 Layout

The dashboard uses a three-column CSS Grid for the skill cards on desktop screens. Below 900px the grid collapses to two columns, and below 600px to a single column. The filter bar uses Flexbox to lay out the filter buttons in a wrapping row.

The summary statistics bar uses a four-column CSS Grid with vertical divider lines between stat blocks. The category legend uses a Flexbox row with wrapping enabled for narrow screens.

The header uses a two-column CSS Grid - title block on the left, status badge on the right - mirroring the resume page's hero structure for visual consistency.

2.5 Interactive Features

1. Custom Cursor

Identical implementation to the resume page - two-part cursor with instant dot and lagging ring, scaling up on hover over skill cards and filter buttons.

2. Category Filter Buttons

Seven buttons above the skill grid allow the user to filter cards by category: All, Security, Networking, Systems, Scripting, Support, and Tools. Clicking a button adds the `.hidden` class to all non-matching cards (hiding them via `display: none`) and removes `.hidden` from matching cards. Cards that are hidden are also unflipped so they reset to their front face. The active button is highlighted with a filled accent background and bright border. Event listeners are attached individually to each button.

3. Card Flip on Click

Clicking any skill card toggles the `.flipped` class on it. The CSS applies a `3D rotateY(180deg)` transform to the `.card-inner` element inside the card. Both faces use `backface-visibility: hidden` so only one face is visible at a time. The front face shows the progress bar and brief description. The back face shows a proficiency level badge (Beginner / Intermediate / Advanced), a paragraph of context about where the skill was applied, a row of related tool tags, and a next-steps goal line. The flip transition is a 0.6-second cubic-bezier ease.

4. Animated Stat Counters

When the summary statistics bar scrolls into view, each stat value counts up from 0 to its target number using a `setInterval` loop. The count animation runs over 1200ms in 40 incremental steps. Each counter reads its target value from a data-

target HTML attribute and an optional data-suffix for units like the % symbol. The IntersectionObserver disconnects after triggering once so the counters only animate on first scroll-in.

5. Scroll-Triggered Progress Bars

An IntersectionObserver on the skills grid triggers all card progress bars to animate from 0% to their data-level target width when the grid first enters the viewport, using the same cubic-bezier transition as the resume page.

6. Scroll Reveal

All skill cards and the legend section use the same .reveal / .visible IntersectionObserver pattern as the resume page.

7. Animated Background Canvas

Identical node network canvas animation to the resume page, providing visual consistency and reinforcing the networking theme.

2.6 Animation Choices

The card flip was chosen as the primary interaction because it gives each skill card two layers of information density without cluttering the initial view. The front provides a quick visual scan of all skills. The back rewards engagement with richer context that would be overwhelming if displayed by default.

The stat counter animation in the summary bar was chosen to draw attention to aggregate achievements in a way that feels dynamic and earned - numbers that count up to their values feel more significant than numbers that simply appear.

2.7 Progress Indicator Style

Skills use a thin 3px horizontal progress bar directly below the skill name on the card front face. The bar is slightly thicker than the resume page bars (3px vs 2px) to be more visible at the smaller card scale. The percentage value is displayed numerically above the bar in a large font (16px) to ensure the exact figure is immediately readable without requiring the user to estimate from the bar length. Bars are colour-coded by category using the same colour system as the category labels.

```
1  :root {
2    --bg-base:      #08080f;
3    --accent:       #4f7cff;
4    --text-primary: #e8e8f5;
5    --border:       rgba(79, 124, 255, 0.14);
6  }

8  .hero-logo {
9    font-family: var(--serif);
10   font-size: clamp(52px, 9vw, 108px);
11   color: var(--text-primary);
12   letter-spacing: -2px;
13 }

15 .exp-card {
16   border: 1px solid var(--border);
17   background: var(--bg-card);
18   border-radius: 4px;
19   overflow: hidden;
20   transition: border-color 0.3s ease;
21 }

23 /* skill bar fill animation */
24 .skill-bar {
25   width: 0%;
```

Midway Build – CSS custom properties and card styles being written

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Technology evolves quickly. Responsibility does not.

EXPERIENCE

IT Support & Cybersecurity Analyst
SECURELOGIX
Jan 2024 – Jun 2025

Cybersecurity Intern
TOOL-GO INC.

Midway Browser Preview – Header and experience cards rendering, skills and contact pending

PROFICIENCY OVERVIEW · 2026

SKILLS DASHBOARD

9 SKILLS TRACKED

Rayyan Umair · IT Support & Cybersecurity

Click any card to see details

FILTER: [ALL] [SECURITY] [NETWORKING] [SYSTEMS] [SCRIPTING] [SUPPORT] [TOOLS]

Skill Cards

SECURITY

CompTIA Security+

Threat analysis, cryptography, access control, and incident response fundamentals.

PROFICIENCY 82%

CLICK TO EXPAND

SECURITY

SIEM Monitoring

Alert review, log analysis, and indicator of compromise identification in live environments.

PROFICIENCY 75%

CLICK TO EXPAND

SECURITY

Penetration Testing

Assisted vulnerability assessments across web and network environments. Supervised engagements at Tool-Go Inc.

PROFICIENCY 70%

CLICK TO EXPAND

3. Tools Used

The following tools were used in the development of both projects:

Development Environment

- Visual Studio Code - primary code editor for all HTML, CSS, and JavaScript files.
- Live Server extension (VS Code) - used for local browser previewing during development.
- Google Chrome Developer Tools - used for inspecting layout, debugging JavaScript, and testing responsive behaviour.

Languages & Technologies

- HTML5 - semantic page structure, accessibility attributes (aria-label, aria-expanded, role).
- CSS3 - custom properties, CSS Grid, Flexbox, transitions, animations, 3D transforms, backdrop-filter.
- Vanilla JavaScript (ES5/ES6) - DOM manipulation, event listeners, IntersectionObserver, Canvas API, requestAnimationFrame.

Design Tools

- No external design tools were used. The visual design was developed directly in CSS using a custom colour palette defined as CSS custom properties.

Version Control

- Git and GitHub Desktop - for version control and submission via GitHub repository.

4. Development Steps

Both projects were developed following a structured methodology. The steps below describe the process for resume.html; dashboard.html followed the same sequence.

Step 1 - Plan Structure and Content

Before writing any code, the content was planned: which sections to include, what data each section would contain, and what interactive features were required to meet the assignment rubric. A consistent visual identity and colour palette was established first so all design decisions could reference it.

Step 2 - Write Semantic HTML Structure

The HTML file was created with the full page structure: DOCTYPE declaration, head block with the comment header, and all semantic section elements. Initial placeholder text was placed inside each section. The HTML was kept completely unstyled at this stage to ensure the structure was sound before any CSS was added.

Step 3 - Initial Commit

The base HTML file with comment block was committed to GitHub as the first commit, establishing the starting point of the commit history.

Step 4 - Add HTML Comments

Descriptive HTML comments were added to every element block - explaining the purpose of each section, what the JS would control, and how each interactive element functioned. This was committed as the second commit.

Step 5 - Write External CSS

The CSS file was written starting with the CSS custom properties (colour palette and fonts), then the reset block, then each section from top to bottom. CSS Grid and Flexbox layouts were built first, then typography, then card styles, then animations and transitions. The page was previewed in the browser at each stage using VS Code Live Server.

Step 6 - Write External JavaScript

The JavaScript file was written function by function in the order they are called: initCursor, initLoader, initCanvas, initAccordions, initScrollReveal, initSkillBars, initNavHighlight. Each function was tested individually in the browser console before moving to the next.

Step 7 - Test and Refine

The completed page was tested in Chrome at multiple screen widths using the browser's responsive design mode. Accessibility was checked by tabbing through all interactive elements to confirm keyboard navigation worked. The HTML and CSS were validated using the W3C Validator.

Step 8 - Final Commit and Push

Final versions of all three files were committed with a final commit message and pushed to the GitHub repository using GitHub Desktop.

5. Challenges Faced

Challenge 1 - fetch() Blocked on file:// URLs

The original version of resume.html used the JavaScript fetch() API to load content from JSON data files. This worked correctly when served by a local web server but failed silently when the HTML file was opened directly by double-clicking it, because browsers block fetch() requests on file:// protocol URLs for security reasons. The solution was to rebuild the page with all content hardcoded directly in the HTML, eliminating the need for any server or data fetching. This also made the file simpler and more reliable for submission.

Challenge 2 - CSS 3D Card Flip on Safari

During testing, the card flip animation on the dashboard worked correctly in Chrome but showed both card faces simultaneously on some browsers due to inconsistent support for backface-visibility. The fix was to add the -webkit-backface-visibility: hidden prefix alongside the standard property on both card faces, ensuring consistent behaviour across browsers.

Challenge 3 - Accordion Max-Height Animation

The accordion expand/collapse animation on the experience cards required animating from max-height: 0 to max-height: auto. CSS cannot transition to auto values, so a fixed max-height: 600px was used for the open state instead. This creates a smooth animation in practice as long as the content does not exceed the fixed maximum height.

6. Sources & Citations

The following sources were consulted and referenced during development:

MDN Web Docs

- HTML elements and semantic structure: <https://developer.mozilla.org/en-US/docs/Web/HTML/Element>
- CSS Grid Layout: https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Grid_Layout
- CSS Flexbox: https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Flexible_Box_Layout
- CSS Custom Properties: https://developer.mozilla.org/en-US/docs/Web/CSS/Using_CSS_custom_properties
- CSS Transitions: <https://developer.mozilla.org/en-US/docs/Web/CSS/transition>
- CSS 3D Transforms (card flip): <https://developer.mozilla.org/en-US/docs/Web/CSS/transform>
- CSS backface-visibility: <https://developer.mozilla.org/en-US/docs/Web/CSS/backface-visibility>
- IntersectionObserver API: https://developer.mozilla.org/en-US/docs/Web/API/Intersection_Observer_API
- Canvas API: https://developer.mozilla.org/en-US/docs/Web/API/Canvas_API
- requestAnimationFrame: <https://developer.mozilla.org/en-US/docs/Web/API/window/requestAnimationFrame>

- MouseEvent: <https://developer.mozilla.org/en-US/docs/Web/API/MouseEvent>
- ARIA attributes: <https://developer.mozilla.org/en-US/docs/Web/Accessibility/ARIA>

W3Schools

- CSS Grid reference: https://www.w3schools.com/css/css_grid.asp
- CSS Flexbox reference: https://www.w3schools.com/css/css3_flexbox.asp
- CSS Animations: https://www.w3schools.com/css/css3_animations.asp
- JavaScript setInterval:
https://www.w3schools.com/jsref/met_win_setinterval.asp

Assignment Reference Material

- INFT1206 Course Notes and Lecture Slides - provided via DC Connect
- MDN JavaScript First Steps: https://developer.mozilla.org/en-US/docs/Learn/JavaScript/First_steps
- MDN JavaScript Building Blocks: https://developer.mozilla.org/en-US/docs/Learn/JavaScript/Building_blocks
- MDN Introducing JavaScript Objects: <https://developer.mozilla.org/en-US/docs/Learn/JavaScript/Objects>